Lecture 7

We will:

1. Revise the concept of a power series and calculate power series for some simple functions
2. Motivate the use of power series for solving ODEs
3. Revise the ratio test for determining the radius of convergence of a power series

Topic 2: Power Series Solutions to ODEs

- You are familiar with the concept of a series, e.g.,

$$\sum_{n=0}^{\infty} c_n$$

  how to test if they converge and what they converge to.

- The series may be functions of t, e.g., a power series

$$\sum_{n=0}^{\infty} c_n (t-a)^n$$

Definition (Taylor series)

- If $f(t)$ is N-times differentiable, the Nth-order Taylor polynomial around $t = a$ is

$$P_N(t) = \sum_{n=0}^{N} \frac{f^{(n)}(a)}{n!} (t-a)^n$$

- If $f(t)$ is infinitely differentiable, then the Taylor series of f is

$$P(t) = \lim_{N\to\infty} P_N(t) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (t-a)^n$$

- If $a = 0$ then it is often known as a Maclaurin series

Exercise 2.1: Find the Taylor series of $f(t) = \mathrm{e}^t$ around $t = 0$

The derivatives are

$$f^{(n)}(t) = \mathrm{e}^t \quad \text{for} \quad n = 0, 1, 2, \ldots$$

$$\Rightarrow \quad f^{(n)}(0) = 1 \quad \text{for} \quad n = 0, 1, 2, \ldots$$

Thus, the Taylor polynomial for e^t around $t = 0$ is

$$P(t) = \sum_{n=0}^{\infty} \frac{t^n}{n!} = 1 + t + \frac{1}{2}t^2 + \ldots$$

Exercise 2.2: Find the Taylor series of $f(t) = \dfrac{1}{1-t}$ about $t = 0$

$$f(t) = \frac{1}{1-t} \qquad \Rightarrow \qquad f(0) = 1$$

$$f'(t) = \frac{1}{(1-t)^2} \qquad \Rightarrow \qquad f'(0) = 1$$

$$f''(t) = \frac{2}{(1-t)^3} \qquad \Rightarrow \qquad f''(0) = 2$$

$$f'''(t) = \frac{6}{(1-t)^4} \qquad \Rightarrow \qquad f'''(0) = 6$$

$$f^{(n)}(t) = \frac{n!}{(1-t)^{n+1}} \qquad \Rightarrow \qquad f^{(n)}(0) = n!$$

Use this to find the Taylor series

$$P(t) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}(t-0)^n = \sum_{n=0}^{\infty} \frac{n!}{n!} t^n = \sum_{n=0}^{\infty} t^n$$

As this is a geometric series it converges for $|t| < 1$

Exercise 2.3: Solve y' = y

The answer is $y(t) = C\,\mathrm{e}^t \ldots \mathrm{buuttt}_{\mathrm{ttt}_{\mathrm{t}}}$

...what if the exponential function hadn't been discovered?

Try a Maclaurin series of form

$$y(t) = \sum_{n=0}^{\infty} c_n\, t^n$$

$$\Rightarrow \quad y'(t) = \sum_{n=1}^{\infty} n\, c_n\, t^{n-1} = \sum_{m=0}^{\infty} (m+1)\, c_{m+1}\, t^m \quad \text{making an assumption!}$$

$$\Rightarrow \quad \underbrace{\sum_{m=0}^{\infty} (m+1)\, c_{m+1}\, t^m}_{y'} = \underbrace{\sum_{m=0}^{\infty} c_m\, t^m}_{y} \quad \Rightarrow \quad \sum_{m=0}^{\infty} \Big\{ (m+1)\, c_{m+1} - c_m \Big\}\, t^m = 0$$

Therefore, require

$$(m+1)\,c_{m+1} - c_m = 0 \quad \text{for} \quad m = 0, 1, 2, \ldots$$

$$c_{m+1} = \frac{c_m}{m+1} \quad \text{for} \quad m = 0, 1, 2, \ldots$$

a <u>recurrence relation</u>.

Let $c_0 = C$, then

$$c_1 = \frac{C}{1}, \quad c_2 = \frac{c_1}{2} = \frac{C}{2 \cdot 1}, \quad c_3 = \frac{c_2}{3} = \frac{C}{3 \cdot 2 \cdot 1}, \ldots \quad c_n = \frac{C}{n!}$$

Our solution is

$$y(t) = \sum_{n=0}^{\infty} c_n\, t^n = C \sum_{n=0}^{\infty} \frac{t^n}{n!} = C\, \mathrm{e}^t$$

Convergence of power series

Theorem (Ratio test for power series)

- Consider the power series
$$P(t) = \sum_{n=0}^{\infty} c_n (t-a)^n$$
where $c_n \neq 0$ for all large n.

- The power series converges if
$$L = \lim_{n\to\infty} \left| \frac{c_{n+1}(t-a)^{n+1}}{c_n(t-a)^n} \right| = \lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n}(t-a) \right| < 1$$
This gives the *radius of convergence*.

Need to use other methods for when $L = 1$ as the ratio test is inconclusive for this case.

Exercise 2.4: For what values of t does the series $\sum_{n=1}^{\infty} \frac{t^n}{2n}$ converge?

We have

$$a = 0 \quad \text{and} \quad c_n = \frac{1}{2n}$$

Apply ratio test

$$L = \lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} t \right| = \lim_{n\to\infty} \left| \frac{2n}{2(n+1)} t \right| = \lim_{n\to\infty} \left(\frac{1}{1+\frac{1}{n}} \right) |t| = |t| < 1$$

Thus, the radius of convergence is one

Test $\sum_{n=1}^{\infty} \frac{t^n}{2n}$ for cases when $L = 1$.

- $t = 1$: leads to $\sum_{n=1}^{\infty} \frac{1}{2n}$, which diverges (recall this is the harmonic $p = 1$ series)
- $t = -1$: this leads to $\sum_{n=1}^{\infty} \frac{(-1)^n}{2n}$ which converges (by the alternating series test)

Therefore, the series converges for $-1 \leqslant t < 1$.

Exercise 2.5: Use the ratio test on the series

$$P(t) = \sum_{n=1}^{\infty} c_n \, t^n \quad \text{where} \quad c_n = \begin{cases} \frac{1}{2^n} & (n \text{ even}) \\ 0 & (n \text{ odd}) \end{cases}$$

i.e., every other term is zero.

Apply ratio test to sequential non-zero terms $n = 2\,k$ and $2\,(k+1)$

$$\lim_{k \to \infty} \left| \frac{t^{2(k+1)}}{2^{2(k+1)}} \frac{2^{2k}}{t^{2k}} \right| < 1$$

$$\Rightarrow \quad \left| \frac{t^2}{4} \right| < 1 \quad \Rightarrow \quad |t| < 2$$

Could also have rewritten the series as $P(t) = \displaystyle\sum_{k=1}^{\infty} \left(\frac{t}{2} \right)^{2k}$

Now testing the endpoints

- Note that that $P(-2) = P(2)$
- $P(2) = \sum_{k=1}^{\infty} \frac{2^{2k}}{2^{2k}} = \sum_{k=1}^{\infty} 1$, which diverges

Thus, $P(t)$ converges for $-2 < t < 2$.

Lecture 7 recap

In Lecture 7:

- Revised power series, including Taylor series representations of functions.
- Saw an example of how to solve an ODE using a power series.
- Revised the ratio test to determine the radius of convergence of a series.

Lecture 8

In Lecture 8, we will:

1. Continue investigating convergence of power series
2. Learn the power series solution methodology
3. Solve the Airy differential equation

Exercise 2.6: What does a given power series converge to?

For what t does the Maclaurin series of $f(t) = \mathrm{e}^t$ converge?

From previous lecture, the Maclaurin series is $P(t) = \sum_{n=0}^{\infty} \frac{t^n}{n!}$.

Use the ratio test to calculate the radius of convergence

$$\lim_{n\to\infty} \left| \frac{n!\,t}{(n+1)!} \right| = \lim_{n\to\infty} \left| \frac{t}{n+1} \right| = 0$$

and $0 < 1$ independent of t

Therefore, the series converges for all t (infinite radius of convergence).

What does it converge to?

Approach: Show the power series satisfies an IVP with a known unique solution.

Differentiate the series (term by term) to give

$$P'(t) = \sum_{n=0}^{\infty} \frac{t^{n-1}n}{n!} = 0 + \sum_{n=1}^{\infty} \frac{t^{n-1}}{(n-1)!} = \sum_{m=0}^{\infty} \frac{t^m}{m!} = P(t)$$

Also, note that

$$P(0) = 1 + \sum_{n=1}^{\infty} \frac{0^n}{n!} = 1 + 0 + \frac{0^2}{2!} + \ldots = 1$$

Thus, we have the IVP

$$P'(t) - P(t) = 0 \quad \text{with} \quad P(0) = 1$$

From Topic 1, we know the IVP has a unique solution, which is $P(t) = \mathrm{e}^t$.

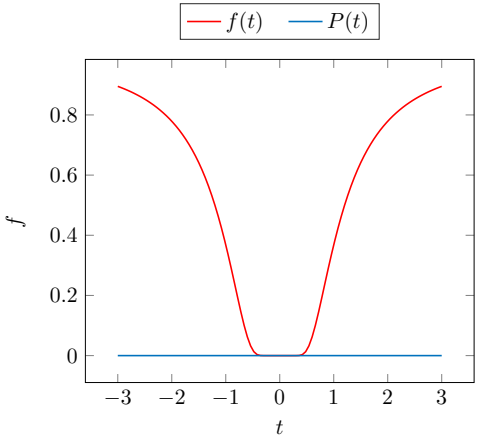
Thus, we have the limit of the series.

Exercise 2.7: Taylor series don't always converge to the original function

Consider the function $f(t) = \begin{cases} e^{-\frac{1}{t^2}} & t \neq 0 \\ 0 & t = 0 \end{cases}$

- $f(t)$ is continuous and infinitely differentiable for all t
- $f^{(n)}(0) = 0$ for all n
- Thus, the Taylor series of $f(t)$ is

$$P(t) = \sum_{n=0}^{\infty} \frac{0}{n!} t^n = 0 \quad \text{for all } t$$



Power series solution methodology

1. Substitute $y(t) = \sum_{n=0}^{\infty} c_n t^n$ into ODE.
2. Change index in each sum to have the same power of t.
3. Extract any low index terms so all sums are over the same range.
4. Solve for the coefficients using recurrence relations.
5. Check if you have the right number of linearly independent solutions.

Exercise 2.8: Find series solutions of $y''(t) + y(t) = 0$

We know two linearly independent solutions are $\cos(t)$ and $\sin(t)$.

Substituting $y(t) = \sum_{n=0}^{\infty} c_n t^n$ into the ODE gives

$$\sum_{n=0}^{\infty} c_n n(n-1)t^{n-2} + \sum_{n=0}^{\infty} c_n t^n = 0$$

In the first sum let $m = n - 2$ and in the second let $m = n$, then

$$\underbrace{\sum_{m=-2}^{\infty} c_{m+2}(m+2)(m+1)t^m}_{= 0 \text{ for } m = -2, -1} + \sum_{m=0}^{\infty} c_m t^m = 0$$

$$\sum_{m=0}^{\infty} \{c_{m+2}(m+2)(m+1) + c_m\}\, t^m = 0$$

We require

$$c_{m+2}(m+2)(m+1) + c_m = 0 \quad \Rightarrow \quad c_{m+2} = \frac{-c_m}{(m+2)(m+1)} \quad \text{for} \quad m = 0, 1, 2, \ldots$$

a recurrence relation.

$$c_2 = \frac{-c_0}{2}, \quad c_4 = \frac{-c_2}{4 \cdot 3} = \frac{c_0}{4 \cdot 3 \cdot 2}, \ldots, \quad c_{2n} = \frac{(-1)^n c_0}{(2n)!}$$

$$c_3 = \frac{-c_1}{3 \cdot 2}, \quad c_5 = \frac{-c_3}{5 \cdot 4} = \frac{c_1}{5 \cdot 4 \cdot 3 \cdot 2}, \ldots, \quad c_{2n+1} = \frac{(-1)^n c_1}{(2n+1)!}$$

We have found two independent series solutions

$$c_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} t^{2n} = c_0 \underbrace{\left(1 - \frac{t^2}{2!} + \dots\right)}_{= \cos(t)}$$

$$\text{and} \quad c_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} t^{2n+1} = c_1 \underbrace{\left(t - \frac{t^3}{3!} + \dots\right)}_{= \sin(t)}$$

Replace c_0 and c_1 with C_1 and C_2, then the general solution is

$$\begin{aligned} y(t) &= C_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} t^{2n} + C_2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} t^{2n+1} \\ &= C_1 \cos(t) + C_2 \sin(t) \end{aligned}$$

A more challenging problem

Airy differential equation

- Is the 2nd-order, homogeneous linear ODE

$$y''(x) - x\,y(x) = 0$$

- It arises when studying optical phenomena, among other things.
- No solutions in terms of elementary functions.

Search for a power series solution: $y(x) = \sum_{n=0}^{\infty} c_n x^n$

$$\sum_{n=0}^{\infty} c_n n(n-1)x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

Use $n - 2 \to m + 1$ in first sum

$$\Rightarrow \quad \sum_{m=-3}^{\infty} c_{m+3}(m+3)(m+2)x^{m+1} - \sum_{m=0}^{\infty} c_m x^{m+1} = 0$$

$$\Rightarrow \quad 2\,c_2 + \sum_{m=0}^{\infty} \{c_{m+3}(m+3)(m+2) - c_m\}\, x^{m+1} = 0$$

$$\Rightarrow \quad c_2 = 0 \qquad \text{and} \qquad c_{m+3} = \frac{c_m}{(m+3)(m+2)}$$

$$c_3 = \frac{1}{2\cdot 3}c_0, \quad \ldots, \quad c_{3k} = \frac{1}{2\cdot 3\cdot 5\cdot 6\cdots(3k-1)\cdot 3k}c_0$$

$$c_4 = \frac{1}{3\cdot 4}c_1, \quad \ldots, \quad c_{3k+1} = \frac{1}{3\cdot 4\cdot 6\cdot 7\cdots 3k\cdot(3k+1)}c_1$$

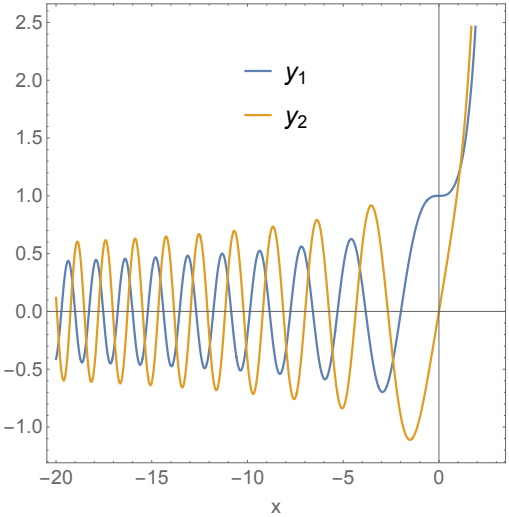
Two independent solutions:

$$y_1(x) = 1 + \sum_{k=1}^{\infty} \frac{x^{3k}}{2 \cdot 3 \cdot 5 \cdot 6 \cdots (3k-1) \cdot 3k}$$

$$y_2(x) = x + \sum_{k=1}^{\infty} \frac{x^{3k+1}}{3 \cdot 4 \cdot 6 \cdot 7 \cdots 3k \cdot (3k+1)}$$

The general solution of Airy's eqn.

$$y(x) = C_1\, y_1(x) + C_2\, y_2(x)$$



Airy functions

The *Airy functions* are

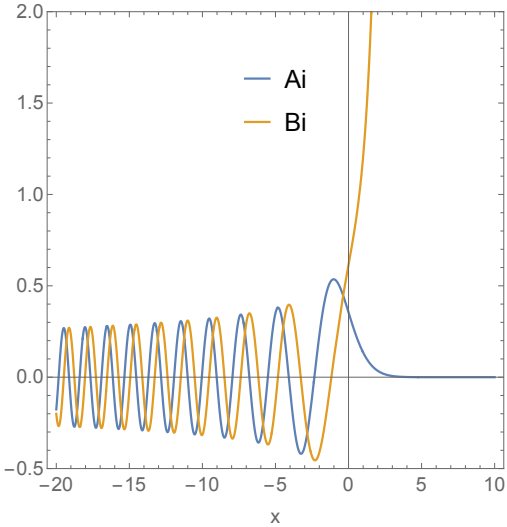
$$\mathrm{Ai}(x) = \alpha\left[y_1(x) - \beta y_2(x)\right]$$

$$\mathrm{Bi}(x) = \sqrt{3}\alpha\left[y_1(x) + \beta y_2(x)\right]$$

where $\alpha \approx 0.355$ and $\beta \approx 0.729$.

Note Ai(x) $\to 0$ as $x \to \infty$

We'll see $\mathrm{Ai}(x)$ again in Topic 6.



Lecture 8 recap

In Lecture 8:

- Learnt a five-step process to calculate power series solutions to ODEs
  1. Substitute power series into ODE
  2. Match powers of t
  3. Make infinite sums over same range
  4. Form recurrence relation
  5. Check number of linearly independent solutions
- Applied method to example problems

Lecture 9

In Lecture 9, we will:

1. Find a power series solution to Bessel's equation
2. Learn the importance of uniform convergence and the Weierstrass test
3. Define *formal* and *genuine* power series solutions

Case study 3: Transmission lines and waveguides

- We see electricity transmission or power lines everywhere everyday, but what are waveguides? Well they are basically devices that 'guide' waves, like sound, transmitting them in one direction with minimal loss of energy.
- They are typically hollow and have a rectangular or circular cross section. Modelling the latter leads to a **differential equation that can be solved using the power series method. The solutions are expressed in terms of so-called Bessel's functions.** (Gómez-Correa et al (2017) American Journal of Physics)

Bessel differential equation

Order zero Bessel equation

- Another 2nd-order, homogeneous linear ODE

$$r^2 y''(r) + r y'(r) + r^2 y(r) = 0$$

- Arises when studying systems with cylindrical or spherical symmetry
- No solutions in terms of elementary functions (similar to Airy equation)

- Bessel's equation in standard form is

$$y''(r) + \frac{1}{r} y'(r) + y(r) = 0.$$

- $1/r$ is NOT continuous on any interval that includes $r = 0$
- ODE <u>does not satisfy</u> requirements for existence and uniqueness of solutions

Exercise 2.8: Find power series solution to Bessel eqn of order zero

Substitute power series of form $y(r) = \sum_{n=0}^{\infty} c_n r^n$ into ODE (standard form)

$$\sum_{n=0}^{\infty} c_n n(n-1) r^{n-2} + \sum_{n=0}^{\infty} c_n n \frac{1}{r} r^{n-1} + \sum_{n=0}^{\infty} c_n r^n = 0$$

$$\Rightarrow \quad \sum_{m=-2}^{\infty} c_{m+2}(m+2)(m+1) r^m + \sum_{m=-2}^{\infty} c_{m+2}(m+2) r^m + \sum_{m=0}^{\infty} c_m r^m = 0$$

$$\Rightarrow \quad \frac{c_1}{r} + \sum_{m=0}^{\infty} \left\{ c_{m+2}(m+2)^2 + c_m \right\} r^m = 0$$

Therefore, $c_1 = 0$ and $c_{m+2} = -\dfrac{c_m}{(m+2)^2}$ for $m \geqslant 0$.

$$c_2 = -\frac{c_0}{2^2}, \quad c_4 = -\frac{c_2}{4^2} = \frac{c_0}{2^2 \cdot 4^2}, \ldots, \quad c_{2k} = \frac{(-1)^k c_0}{2^2 \cdot 4^2 \ldots (2k)^2} = \frac{(-1)^k c_0}{(2^k k!)^2}$$

$$c_3 = -\frac{c_1}{3^2} = 0, \quad c_5 = 0, \ldots, \quad c_{2k+1} = 0$$

The power series method only found one linearly independent solution, which is

$$y_1(r) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2^k k!)^2} r^{2k} = 1 - \frac{r^2}{4} + \ldots \equiv J_0(r)$$

J_0 is the <u>Bessel function of the first kind of order zero</u>.

Exercise 2.9: Find general solution of Bessel eqn of order zero

Use reduction of order $y(r) = u(r)\,v(r)$ with $v = J_0(r)$

For the Bessel eqn $f_1(r) = 1/r$, and so the reduction of order integrating factor is

$$I = v^2 \mathrm{e}^{\int \frac{1}{r}\mathrm{d}r} = r\,v^2$$

Reduction of order requires we find $w = u'$ from

$$\frac{\mathrm{d}}{\mathrm{d}r}[I\,w] = \frac{I\,g}{v} \qquad \Rightarrow \qquad r\,v^2 u' = C_2 \qquad \Rightarrow \qquad u = C_1 + \int \frac{C_2}{r v^2}$$

We can't evaluate the integral, and so we just write

$$y(r) = C_1\,J_0(r) + C_2\,\underbrace{J_0(r)\int \frac{1}{r J_0^2}}_{y_2(r)}$$

Differentiating under the summation

- The method relies on differentiating the power series term by term
- Is this valid?

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_{n=0}^{\infty} c_n t^n \stackrel{?}{=} \sum_{n=0}^{\infty} \frac{\mathrm{d}}{\mathrm{d}t} c_n t^n$$

Definition (Uniform convergence)

Sequence $W_N(t)$ converges *uniformly* to $W(t)$ on $t \in D$ if

$$\lim_{N\to\infty} \max_{t\in D} |W_N(t) - W(t)| = 0$$

This means the *maximum* error goes to zero as $N \to \infty$.

Stronger condition than pointwise convergence.

Strictly, max should be sup, which you will see if you take MAST20026 Real Analysis.

Definition (Uniform convergence of a series)

- Let $W_N(t) = \sum_{n=1}^{N} w_n(t)$ converge uniformly to $W(t)$ on D
- Then $\sum_{n=1}^{\infty} w_n(t)$ converges uniformly to $W(t)$ on D, i.e.,

$$\lim_{N\to\infty} \max_{t\in D} \left| \sum_{n=1}^{N} w_n(t) - W(t) \right| = 0$$

Theorem (Weierstrass test for uniform convergence)

- Let $w_n(t)$ $(n = 1, 2, 3, \ldots)$ be functions on D
- Let $\max_{t\in D} |w_n(t)| \leqslant a_n$ for some $a_n \in \mathbb{R}$ such that $\sum_{n=1}^{\infty} a_n$ converges
- Then $\sum_{n=1}^{\infty} w_n(t)$ converges *uniformly* on D

Exercise 2.10: Does $\displaystyle\sum_{n=0}^{\infty} \frac{t^n}{n!}$ converge uniformly on $t \in [-b, b]$?

Let $a_n = \dfrac{b^n}{n!}$ and so $\displaystyle\max_{t \in D} \left| \frac{t^n}{n!} \right| \leqslant a_n$

$\displaystyle\sum_{n=0}^{\infty} a_n$ converges by the ratio test:

$$\lim_{n \to \infty} \left| \frac{b^{n+1}}{(n+1)!} \frac{n!}{b^n} \right| = \lim_{n \to \infty} \left| \frac{\frac{b}{n}}{1 + \frac{1}{n}} \right| = 0 < 1$$

Therefore, $\displaystyle\sum_{n=0}^{\infty} \frac{t^n}{n!}$ converges uniformly on D by the Weierstrass test

- Let $P(t) = \sum_{n=0}^{\infty} w_n(t)$ where w_n are continuous differentiable functions on $D = [a, b]$
- Suppose $P(t)$ converges uniformly on D
- Suppose $\sum_{n=0}^{\infty} w_n'(t)$ converges uniformly on D

Theorem (Differentiating a series)

Then $P(t)$ can be differentiated term by term, i.e.,

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t} \sum_{n=0}^{\infty} w_n(t) = \sum_{n=0}^{\infty} \frac{\mathrm{d}}{\mathrm{d}t} w_n(t)$$

Theorem (Uniform convergence of a power series)

Every power series converges uniformly on every closed interval that lies strictly within its interval of convergence.

Formal and genuine solutions

1. Formal solution: term by term differentiated series satisfies the ODE.
   - Series and its derivatives may or may not converge (either pointwise or uniformly).
   - Series solution may or may not satisfy the ODE.
2. Genuine solution: series and all derivatives up to order of ODE converge uniformly.
   - Term by term differentiation up to the order of the ODE is allowed.
   - Solution satisfies the ODE.

Power series and uniform convergence

Search for a *formal* power series solution to an ODE

$$y(t) = \sum_{n=0}^{\infty} c_n t^n$$

and find coefficients so that the term by term series satisfies the ODE.

If the radius of convergence is non-zero then, for any domain strictly within the radius of convergence, the power series

1. converges uniformly,
2. can be differentiated term by term and
3. is a *genuine solution* to the ODE.